# Use Case Template

## CardioAI - Heart Disease Prediction System

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| **1. Use Case Title** | Compute SHAP Values |
| **2. Abbreviated Title** | Compute SHAP |
| **3. Use Case Id** | UC-13 |
| **4. Actors** | ML Engine (Supporting) |
| **5. Description** | The ML Engine computes SHAP (SHapley Additive exPlanations) values to quantify each of the 13 clinical features' marginal contribution to the model's prediction output. |
| **5.1. Pre Conditions** | 1. Model inference (UC-11) has been completed.<br>2. Background dataset (training sample) is available for SHAP explainer. |
| **5.2. Task Sequence** | 1. System initializes DeepSHAP explainer with background data.<br>2. System computes SHAP values for the current patient input.<br>3. System ranks features by absolute SHAP value.<br>4. System generates positive/negative contribution indicators.<br>5. System outputs sorted feature importance list for visualization. |
| **5.3. Post Conditions** | 1. A vector of 13 SHAP values (one per feature) is produced.<br>2. Feature importance ranking is ready for bar/waterfall chart rendering. |
| **6. Modification History** | Date: September 11, 2026 |
| **7. Author** | Dhruv Singla (1024170434), Srishti (1024170451) |